

## Benchmark

**MA.912.GR.1.2** Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.

### Benchmark Clarifications:

*Clarification 1:* Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.

*Clarification 2:* Instruction focuses on helping a student choose a method they can use reliably.

## Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.2.6, MA.912.GR.2.7, MA.912.GR.2.8, MA.912.GR.2.9
- MA.912.GR.5
- MA.912.T.1.1, MA.912.T.1.2, MA.912.T.1.3, MA.912.T.1.4
- MA.912.LT.4.3, MA.912.LT.4.8, MA.912.LT.4.10

## Terms from the K-12 Glossary

- Angle
- Congruent
- Corresponding Angles
- Hypotenuse
- Right Triangle
- Similarity
- Triangle
- Vertical Angles

## Vertical Alignment

### Previous Benchmarks

- MA.8.GR.1.5
- MA.8.GR.2.1, MA.8.GR.2.2, MA.8.GR.2.4

### Next Benchmarks

- MA.912.NSO.3
- MA.912.T.1.8

## Purpose and Instructional Strategies

In grade 8, students were introduced to the concepts of congruence and similarity using transformations and proportional relationships. In Geometry, students prove triangle congruence or similarity using Side-Side-Side (SSS), Side-Angle-Side (SAS), Angle-Side-Angle (ASA), Angle-Angle-Side (AAS), Angle-Angle (AA) and Hypotenuse-Leg (HL) criteria. In later courses, these congruence and similarity criteria are critical in understanding applications of vectors and trigonometry.

- Students should develop the understanding of which criteria can be used to prove congruence and which criteria should be used to prove similarity.
  - Criteria that can be used to prove congruence are SSS, SAS, ASA, AAS and HL, where the assumption is that the respective sides in the criterion are equal in length, and the respective angles are equal in measure.
  - Criteria that can be used to prove similarity are AA, SAS and SSS, where the assumption is that the respective sides in the criterion are proportional in length with the same constant of proportionality, and the respective angles are equal in measure.
- Instruction includes developing the understanding that congruence implies similarity. Students should realize that if they have proved congruence of two triangles, then they

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have also proved similarity of the two triangles.

- Instruction includes the connection to the Logic and Discrete Theory benchmarks when developing proofs. Additionally, with the construction of proofs, instruction reinforces the Properties of Operations, Equality and Inequality. (*MTR.5.1*)
- Instruction utilizes different ways students can organize their reasoning by constructing various proofs when proving geometric statements. It is important to explain the terms statements and reasons, their roles in a geometric proof, and how they must correspond to each other. Regardless of the style, a geometric proof is a carefully written argument that begins with known facts, proceeds from there through a series of logical deductions, and ends with the statement you are trying to prove. (*MTR.2.1*)
  - For examples of different types of proofs, please see MA.912.LT.4.8.
- Instruction includes the use of hatch marks, hash marks, arc marks or tick marks, a form of mathematical notation, to represent segments of equal length or angles of equal measure in diagrams and images.
- Students should understand the difference between congruent and equal. If two segments are congruent (i.e.,  $\overline{PQ} \cong \overline{MN}$ ), then they have equivalent lengths (i.e.,  $PQ = MN$ ) and the converse is true. If two angles are congruent (i.e.,  $\angle ABC \cong \angle PQR$ ), then they have equivalent angle measure (i.e.,  $m\angle ABC = m\angle PQR$ ) and the converse is true.
- Problem types include mathematical or real-world context where students identify which one of the congruence or similarity criteria can be applied in specific cases; deduce information (e.g., vertical angles are congruent, reflexive property for a shared side or angle, corresponding angles when two parallel lines are cut by a transversal are congruent) from given images to determine the congruence or similarity criterion needed to prove triangle congruence or similarity; and determine what piece of information about a pair of triangles must be added to prove triangles are congruent or similar by a certain criterion. (*MTR.6.1*)
- Instruction includes exploring why Hypotenuse-Leg (HL) can be used to show right triangles are congruent. Students should be able to realize that HL is a specific case of Side-Side-Side (SSS) and Side-Angle-Side (SAS) applying the Pythagorean Theorem.
- Instruction for some theorems, relationships or postulates may be necessary in order to prove the validity of congruence or similarity criteria.
  - For example, to prove the validity of the AA similarity criterion, students will need knowledge of the Triangle Sum Theorem.

### **Common Misconceptions or Errors**

- Students may confuse the congruence and similarity versions of the Side-Side-Side and Side-Angle-Side criteria. To address this misconception, provide students with counterexamples and opportunities to discuss the difference.
- Students may try to use Angle-Angle, Angle-Angle-Angle or Side-Side-Angle to prove congruence.